

Cost analysis question guide

27 April 2025 13:31

A 450g aluminum alloy part can be manufactured by either machining from a solid casting or by precision die casting. The capital costs for both the precision CNC machine and the precision die casting machine are similar so for comparison we have ignored them and the labour costs have also been ignored whilst assuming the following information.

Aluminium alloy material cost - £3.00 /kg

The waste material created per part by machining from a large solid is 650g.

The amount of waste material created per part by die cast moulding is 50g.

The tooling required to machine these parts is £10,000.

The die cast tooling costs dedicated for die casting this part are - £25,000

What is the cost (in £ per part) of machining 1,000 of these parts?

① Determine equation:

$$C = \left[\frac{m C_m}{1-f} \right] + \left[\frac{C_t}{n} \right] + \frac{1}{n} \left[\frac{C_c}{L \cdot t_{wo}} + \dot{C}_{oh} \right]$$

Ignoring capital and labour costs

∴ Equation for cost calculation:

$$C = \left[\frac{m C_m}{1-f} \right] + \left[\frac{C_t}{n} \right]$$

What is the cost (in £ per part) of machining 1,000 of these parts?

Answer: 13.30

② Calculate cost of machining 1000 of these parts:

$$C = \left[\frac{m C_m}{1-f} \right] + \left[\frac{C_t}{n} \right]$$

$$C_m = £3.00/\text{kg}$$

$$m \text{ (mass used per unit)} = 450\text{g} = 0.45\text{kg}$$

$$\text{Waste generated from machining} = 650\text{g} = 0.65\text{kg}$$

$$\text{Scrap fraction, } f = \frac{\text{Waste mass}}{\text{Total mass of material used}} = \frac{0.65}{0.65 + 0.45} = \frac{13}{22} \text{ (0.590...)}$$

$$\text{Cost of tooling (Machining equipment), } C_t = £10,000$$

$$n = 1000$$

$$\therefore \text{Cost per unit (For machining), } C = \left[\frac{0.45 \times 3}{1 - \frac{13}{22}} \right] + \left[\frac{10000}{1000} \right] = 3.3 + 10 = £13.3 \text{ (Per part if 1000 parts are made)}$$

What is the cost (in £ per part) of die casting 1000 of these parts?

Answer: 26.5

③ Calculate cost of Die casting 1000 of these parts:

$$C = \left[\frac{m C_m}{1-f} \right] + \left[\frac{C_t}{n} \right]$$

$$C_m = £3.00/\text{kg}$$

$$m \text{ (mass used per unit)} = 450\text{g} = 0.45\text{kg}$$

$$\text{Waste generated from Die casting} = 50\text{g} = 0.05\text{kg}$$

$$\text{Scrap fraction, } f = \frac{\text{Waste mass}}{\text{Total mass of material used}} = \frac{0.05}{0.05 + 0.45} = 0.1$$

$$\text{Cost of tooling (Die casting equipment), } C_t = £25,000$$

$$n = 1000$$

$$\therefore \text{Cost per unit (For die casting), } C = \left[\frac{0.45 \times 3}{1 - 0.1} \right] + \left[\frac{25000}{1000} \right] = 1.5 + 25 = £26.5 \text{ (Per part if 1000 parts are made)}$$

Examining your previous 2 answers, identify which process you would choose therefore to manufacture just 1,000 parts?

Select one:

- ☒ a. Machining
- ☐ b. Die Casting

The Cost per Unit of Output

Queen Mary
University of London

Material costs C_m , and a mass m is used per unit; f is the scrap fraction (the fraction thrown away)

$$\frac{m C_m}{1-f}$$

Tooling C_t is "dedicated" – it is written off against the number of parts to be made, n

$$\frac{C_t}{n}$$

Capital cost C_c of equipment is "non-dedicated" (it can be used for other products). It is written off against time, giving an hourly rate. The write-off time is t_{wo} . The rate of production is $\dot{n}$ units/hour. The load factor (fraction of time the equipment is used) is L .

$$\frac{1}{\dot{n}} \left[\frac{C_c}{L \cdot t_{wo}} \right]$$

The gross **overhead** rate $\dot{C}_{oh}$ contributes a cost per unit of time that, like capital, depends on production rate $\dot{n}$

$$\frac{\dot{C}_{oh}}{\dot{n}}$$

$$C = \left[\frac{m C_m}{1-f} \right] + \left[\frac{C_t}{n} \right] + \frac{1}{\dot{n}} \left[\frac{C_c}{L \cdot t_{wo}} + \dot{C}_{oh} \right]$$

Materials Tooling Labor, Capital, Information, Energy...
↓ ↓ ↓
Batch size Rate of production

What is the cost (in £ per part) of machining 10,000 of these parts?

Answer: 4.3

④ Calculate cost of machining 10000 of these parts:

$$C = \left[\frac{m C_m}{1-f} \right] + \left[\frac{C_t}{n} \right]$$

$$C_m = £3.00/\text{Kg}$$

$$m \text{ (mass used per unit)} = 450\text{g} = 0.45 \text{ Kg}$$

$$\text{Waste generated from machining} = 650\text{g} = 0.65 \text{ Kg}$$

$$\text{Scrap fraction, } f = \frac{\text{Waste mass}}{\text{Total mass of material used}} = \frac{0.65}{0.65 + 0.45} = \frac{13}{22} (0.590\dots)$$

$$\text{Cost of tooling (Machining equipment), } C_t = £10,000$$

$$n=10000 \rightarrow \text{ONLY } n \text{ CHANGES}$$

$$\therefore \text{Cost per unit (For machining), } C = \left[\frac{0.45 \times 3}{1 - \frac{13}{22}} \right] + \left[\frac{10000}{10000} \right] = 3.3 + 1 = £4.30 \text{ (Per part if 10000 parts are made)}$$

What is the cost (in £ per part) of die casting 10,000 of these parts?

Answer: 4

⑤ Calculate cost of Die casting 10000 of these parts:

$$C = \left[\frac{m C_m}{1-f} \right] + \left[\frac{C_t}{n} \right]$$

$$C_m = £3.00/\text{Kg}$$

$$m \text{ (mass used per unit)} = 450\text{g} = 0.45 \text{ Kg}$$

$$\text{Waste generated from Die casting} = 50\text{g} = 0.05 \text{ Kg}$$

$$\text{Scrap fraction, } f = \frac{\text{Waste mass}}{\text{Total mass of material used}} = \frac{0.05}{0.05 + 0.45} = 0.1$$

$$\text{Cost of tooling (Die casting equipment), } C_t = £25,000$$

$$n=10000 \rightarrow \text{ONLY } n \text{ CHANGES}$$

$$\therefore \text{Cost per unit (For Die casting), } C = \left[\frac{0.45 \times 3}{1 - 0.1} \right] + \left[\frac{25000}{10000} \right] = 1.5 + 2.5 = £4.0 \text{ (Per part if 10000 parts are made)}$$

What manufacturing number is the cross over point where the cost per part is equivalent for both methods of manufacture?

Answer: 8333

⑥ Equate both Cost per unit equations and solve for n :

$$C(\text{machining}) = 3.3 + \frac{10000}{n}$$

$$C(\text{Die casting}) = 1.5 + \frac{25000}{n}$$

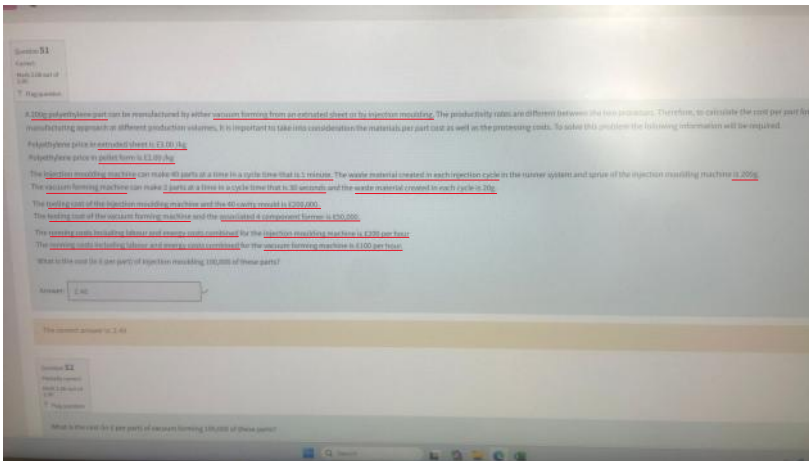
$$C(\text{machining}) = C(\text{Die casting})$$

$$\therefore 3.3 + \frac{10000}{n} = 1.5 + \frac{25000}{n}$$

$$3.3 - 1.5 = \frac{25000}{n} - \frac{10000}{n}$$

$$1.8 = \frac{15000}{n}$$

$$\therefore n = \frac{15000}{1.8} = 8333.3 \text{ units}$$



The Cost per Unit of Output

Material costs C_m , and a mass m is used per unit; f is the scrap fraction (the fraction thrown away)

$$\frac{m C_m}{1-f}$$

Tooling C_t is "dedicated" -- it is written off against the number of parts to be made, n

$$\frac{C_t}{n}$$

Capital cost C_c of equipment is "non-dedicated" (it can be used for other products). It is written off against time, giving an hourly rate. The write-off time is t_{wo} . The rate of production is $\dot{n}$ units/hour. The load factor (fraction of time the equipment is used) is L .

$$\frac{1}{\dot{n}} \left[\frac{C_c}{L \cdot t_{wo}} \right]$$

The gross **overhead rate** $\dot{C}_{oh}$ contributes a cost per unit of time that, like capital, depends on production rate $\dot{n}$

$$\frac{\dot{C}_{oh}}{\dot{n}}$$

$$C = \left[\frac{m C_m}{1-f} \right] + \left[\frac{C_t}{n} \right] + \frac{1}{\dot{n}} \left[\frac{C_c}{L \cdot t_{wo}} + \dot{C}_{oh} \right]$$

Materials Tooling Labor, Capital, Information, Energy...
↓ ↓ ↓
Batch size Rate of production

① Determine equation:

$$C = \left[\frac{m C_m}{1-f} \right] + \left[\frac{C_t}{n} \right] + \frac{1}{\dot{n}} \left[\frac{C_c}{L \cdot t_{wo}} + \dot{C}_{oh} \right]$$

Combined Labour + Energy costs, C_r

Equation for cost calculation:

$$C = \left[\frac{m C_m}{1-f} \right] + \left[\frac{C_t}{n} \right] + \frac{1}{\dot{n}} [C_r]$$

② Calculate cost of injection moulding 100000 of these parts:

$$C = \left[\frac{m C_m}{1-f} \right] + \left[\frac{C_t}{n} \right] + \frac{1}{\dot{n}} [C_r]$$

$$C_m = £2.00/\text{kg}$$

$$m \text{ (mass used per unit)} = 200\text{g} = 0.20 \text{ kg}$$

Waste generated from injection moulding 40 parts = 200g

$$\therefore \text{Waste from one part} = \frac{200}{40} = 5\text{g} = 0.005 \text{ kg}$$

$$\text{Scrap fraction, } f = \frac{\text{Waste mass}}{\text{Total mass of material used}} = \frac{0.005}{0.2 + 0.005} = \frac{1}{41} (0.0244...)$$

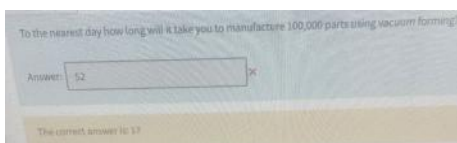
Cost of tooling (Injection moulding equipment), $C_t = £200,000$

$$n = 100000$$

$$\dot{n} = \frac{40 \text{ (parts)}}{60 \text{ (secs)}} = \frac{2}{3} \text{ parts/sec}$$

$$C_r \text{ (Combined running costs)} = £200/\text{hour} = \frac{200}{3600} (£/\text{second})$$

$$\therefore \text{Cost per unit (For machining), } C = \left[\frac{0.20 \times 2}{1 - \frac{1}{41}} \right] + \left[\frac{200000}{100000} \right] + \left[\frac{1}{\left(\frac{2}{3} \right)} \right] \left[\frac{200}{3600} \right] = 0.41 + 2 + 0.0833... \approx £2.493 \text{ per part}$$



Vacuum forming:

$$2 \text{ parts every } 30 \text{ secs} \Rightarrow \frac{2}{30} \text{ parts/sec}$$

Goal = 100000 parts

$$\therefore \text{Time (secs)} = \frac{100000}{\left(\frac{2}{30} \right)} = 1500000 \text{ secs}$$

$$\text{Time (days)} = \frac{1500000}{60 \times 60 \times 24} \approx 17.36 \text{ days}$$